

CARMEL POLYTECHNIC COLLEGE

Virtual Lab Operations — Complete Staff Manual & Step-by-Step Guide

Curriculum Compliance: SBTE Revision 2021 (75 Marks CIA) • All Dates Standardized in DD-MM-YYYY

1. REVISION 2021 MARKS STRUCTURE & ASSESSMENT SPLIT-UP (75M TOTAL CIA)

ASSESSMENT COMPONENT	COMPONENT MAX	CIA WEIGHT	EVALUATION RULES & RUBRICS
Continuous Lab Work	37.5 Marks	37.5 Marks	Average of all conducted experiments evaluated across 5 rubrics (Max 37.5M)
Open-Ended Experiment	7.5 Marks	7.5 Marks	Micro-project / individual problem-solving challenge evaluated out of 7.5 Marks
Practical Series Tests	Test 1: 15M Test 2: 15M	15.0 Marks	Average of Practical Test 1 & Test 2: (Test 1 + Test 2) / 2 (Max 15M)
Lab Attendance	100% Scale	15.0 Marks	Proportional: (Attendance % / 100) × 15 (e.g. 100% → 15M, 90% → 13.5M, 80% → 12M, 75% → 11.3M)
TOTAL CONTINUOUS INTERNAL ASSESSMENT (CIA)		75 MARKS	37.5 (Lab Work) + 7.5 (Open-Ended) + 15 (Tests) + 15 (Attendance) = 75M

5 Rubrics for Continuous Lab Work (37.5 Marks Total per Experiment):

1. **Rough Record / Algorithm:** 0 – 5.0 Marks
 2. **Fair Record / Documentation:** 0 – 7.5 Marks
 3. **Observation / Preparation:** 0 – 7.5 Marks
 4. **Execution / Procedure & Punctuality:** 0 – 7.5 Marks
 5. **Viva Voce / Viva Understanding:** 0 – 10.0 Marks
- Total per experiment = 5 + 7.5 + 7.5 + 7.5 + 10 = 37.5 Marks

2. ROLE DEMARCATION: TUTOR VS ASSIGNED LAB STAFF

ROLE	RESPONSIBILITIES IN VIRTUAL LAB
Tutor / Class In-Charge	Initial student roster verification, mentor assignment, and setting up the Lab Batch Division (Batch 1 / Batch 2 or Whole Class).
Assigned Lab Staff (Faculty In-Charge & Instructors)	Syllabus upload, CO-PO mapping, lesson planning, daily session logging (DD-MM-YYYY), period selection (P1–P3/P4–P6), experiment linking, 5-rubric grading (37.5M), log editing/deleting, date fixes, open-ended evaluations (7.5M), series test question preparation & scoring (15M), threshold configuration, OBE attainment calculation, and official A4 report printing.

3. STEP-BY-STEP OPERATIONS FOR ASSIGNED LAB STAFF (FROM SCRATCH)

Step 1: Login & Navigation

- **Desktop:** Log in with faculty credentials → *Lecturer Dashboard* → Click your assigned Practical Course card (e.g. 3049 - Fundamentals of C Programming Lab).
- **Mobile:** Log in → Tap *Virtual Lab* in bottom menu → Select your Subject.

Step 2: Upload Syllabus & Course Structure Setup

- Open the **Syllabus / Course Structure** tab.
- Upload the official SBTE Revision 2021 syllabus PDF/Word document or import from template.
- Verify course metadata: Subject Name, Subject Code, Semester, Department, Lecture/Lab Hours, and Credits.

Step 3: CO - PO - PSO Setup & Mapping Matrix

- Open **CO-PO Mapping** tab.
- Set up Course Outcomes (CO1, CO2, CO3, etc.) with Bloom's Taxonomy cognitive levels and practical competencies.
- Define correlation levels with Program Outcomes (PO1 to PO7) and Program Specific Outcomes (PSOs):
3 = Substantial (High), 2 = Moderate (Medium), 1 = Slight (Low), - = No Correlation.

Step 4: Lesson Plan Setup & Schedule

- Open the **Lesson Plan / Planner** tab.
- Configure day-wise practical session entries: Day No, Proposed Date (DD-MM-YYYY), Topic/Experiment description, mapped CO Tag, Pedagogy, and Actual Conducted Date.
- Auto-saves as you enter or import from the department template.

Step 5: Formative Tab (Continuous Lab Work & Experiment List)

- **Experiment List Setup:** Verify all syllabus experiments (Exp 1 to Exp 20) with titles and mapped CO tags.
- **Grading Students with 5 Rubric Sliders:**
 - *On Desktop:* Open *Continuous Evaluation* → Select Exp No → Click student row → Adjust the 5 compact horizontal sliders (Rough 5M, Fair 7.5M, Obs 7.5M, Proc 7.5M, Viva 10M) arranged neatly on a single row → Live total (Max 37.5M) updates → Click **Save / Next**.
 - *On Mobile:* Tap student card → Adjust touch sliders → Instant auto-save.
- **Open-Ended Experiment (7.5M):** Enter assigned micro-project topic & evaluated score out of **7.5 Marks**.

Step 6: Summative Tab (Practical Series Tests & Question Bank)

- **Question Bank Preparation:** Add practical questions tagged with COs and max marks into the Course Question Bank.
- **Test Question Setup:** Pick questions from the Question Bank or generate structured question papers for Practical Test 1 & Practical Test 2.
- **Test Mark Entry (15M Total):** Enter exam dates (DD-MM-YYYY) → Enter **Test 1 (Max 15M)** and **Test 2 (Max 15M)**. System automatically computes the average: $(\text{Test 1} + \text{Test 2}) / 2$ (Max 15M).

Step 7: Daily Attendance & Class Logging (Period Presets, Dates & Batches)

- Open **Attendance & Class Log**. Select session date (DD-MM-YYYY).
- Select continuous period preset: **P1-P3 Morning (3h)** or **P4-P6 Afternoon (3h)**.
- Select the batch: **Batch 1**, **Batch 2**, or **Whole Class**.
- Select syllabus experiment from dropdown to auto-link hours and CO tags.
- Click **Mark All Present** (uncheck absentees or tap grid buttons) → Click **Save Class Log & Attendance**.

Step 8: Log Management & Debugging (Edit / Delete / Date Adjustments)

- **Delete Accidental Duplicate Logs:** Open *Past Logs* → Look for amber **Duplicate** badge → Click **Delete**. The duplicate is deleted safely while preserving original logs and attendance.
- **Edit Experiment Conducted Date:** Open *Completed Experiments* modal → Click date picker or **Tune** icon → Set corrected DD-MM-YYYY date → Select scope (*Universal* or *Batch-specific*) → Click **Save & Update**.

Step 9: Proportional Attendance Calculation & Live 75M Master Sheet

- System calculates attendance marks proportionately: $(\text{Present Classes} / \text{Total Classes}) \times 15$.
- Open **Consolidated CIA (75M)** to review the master evaluation sheet:
37.5M (Lab Work) + 7.5M (Open-Ended) + 15M (Series Tests) + 15M (Attendance) = 75.0M TOTAL CIA.
- Faculty can review and fine-tune attendance marks (0–15) in this sheet if institutional duty leaves or approved condonation apply.

4. OBE ATTAINMENT WORKFLOW: SURVEYS, END-SEM GRADES & THRESHOLDS

1. End-Semester Course Exit Survey (Indirect Attainment - 20%):

Students log in to their student portal and complete the Course Exit Survey rating their competence for each Course Outcome (CO) on a 3-point scale:

- 3 (High / Substantial) | 2 (Moderate) | 1 (Slight / Low)

System computes average indirect attainment rating on a 3.0 scale: $\text{Indirect Level} = (\text{Average Score} / 3) \times 3.0$.

2. Threshold Configuration by Faculty (Before Calculation):

- Internal CIA Mark Threshold: e.g. Score $\geq$ 60% or 70% of marks in CO-mapped components.
- End-Semester Examination (ESE) Grade Threshold: e.g. Achieving Grade A or B (or $\geq$ 50% / 60% mark equivalent).
- Student Cohort Target %: Percentage of students achieving threshold to earn Level 3, 2, or 1.

3. End-Semester Exam (ESE) Attainment (Uniform Across All COs):

Because SBTE publishes semester exam results as overall course Grades (S, A, B, C, D, F) rather than individual CO question marks, the End-Semester Exam attainment is uniform across all Course Outcomes (COs) in that course.

Example ESE Attainment (Target: Grade A):

- Level 3 (High) : $\geq$ 70% of total students achieve Grade A or above.
- Level 2 (Medium) : 60% – 69% of students achieve Grade A or above.
- Level 1 (Low) : 50% – 59% of students achieve Grade A or above.
- Level 0 (Unattained) : < 50% of students achieve Grade A or above.

4. Internal CIA Direct Attainment (Per CO):

Computed for each specific CO from its mapped Continuous Lab Work (37.5M), Open-Ended (7.5M), and Series Tests (15M):

- If $\geq$ 70% (or configured target, e.g. 65%) of students score $\geq$ 60% marks in CO1 tasks → CO1 Internal Level = 3.

5. Final Overall CO Attainment Formula (100% Total):

ATTAINMENT COMPONENT	WEIGHTAGE	CALCULATION & DATA SOURCE
Direct Attainment	80%	$(50\% \times \text{Internal CIA Level}) + (50\% \times \text{End-Sem Grade Level})$
Indirect Attainment	20%	Student End-Semester Course Exit Survey average rating (Scale 1 to 3).
FINAL OVERALL CO ATTAINMENT	100%	$(0.80 \times \text{Direct Attainment}) + (0.20 \times \text{Indirect Attainment}) \rightarrow$ Mapped to NBA CO-PO-PSO Matrix

5. MOBILE VIRTUAL LAB WORKFLOW (FLOOR USAGE)

MOBILE FEATURE	OPERATION & USAGE IN LAB FLOOR
Attendance Taking	Quick 1-tap roll call with batch filter and continuous period presets (P1-P3).
Touch Rubric Sliders	Tap student card → Adjust the 5 sliders for Rough, Fair, Obs, Proc, Viva → Real-time instant auto-save.
Log Management	View past logs grouped by date (DD-MM-YYYY) with amber duplicate alert badges & 1-tap deletion.
Student Detail Card	Tap student name to view complete CIA summary breakdown and per-experiment grade cards.

6. OFFICIAL REPORTS & DOCUMENTATION (PRINTABLE A4)

REPORT NAME	LAYOUT	PURPOSE & CONTENT
Practical Experiments Conducted Log	A4 Portrait	Coverage %, actual hours, batch details, conducted dates (DD-MM-YYYY), attendance %.
Consolidated Evaluation Register	A4 Portrait	Full 75M CIA breakdown (37.5M + 7.5M + 15M + 15M) with signatures.
Consolidated Attendance Matrix	A4 Landscape	Date-wise and period-wise presence/absence matrix.
CO-PO-PSO Attainment Report	A4 Portrait	Direct (80%) + Indirect Exit Survey (20%) attainment matrices with threshold audits.

✓ DOs (Best Practices)

- Log continuous 3-hour sessions using multi-period selector (e.g. P1-P3).
- Link syllabus experiment from dropdown to maintain CO tags.

✗ DON'Ts

- Do not create separate 1-hour logs for a continuous 3-hour lab.
- Do not delete student rosters manually (handled by Tutor).
- Do not leave experiments unlinked to Course Outcomes.

- Grade rubric sliders while observing student in lab.
- Configure attainment thresholds prior to final NBA report export.

- Do not print registers without verifying DD-MM-YYYY dates.